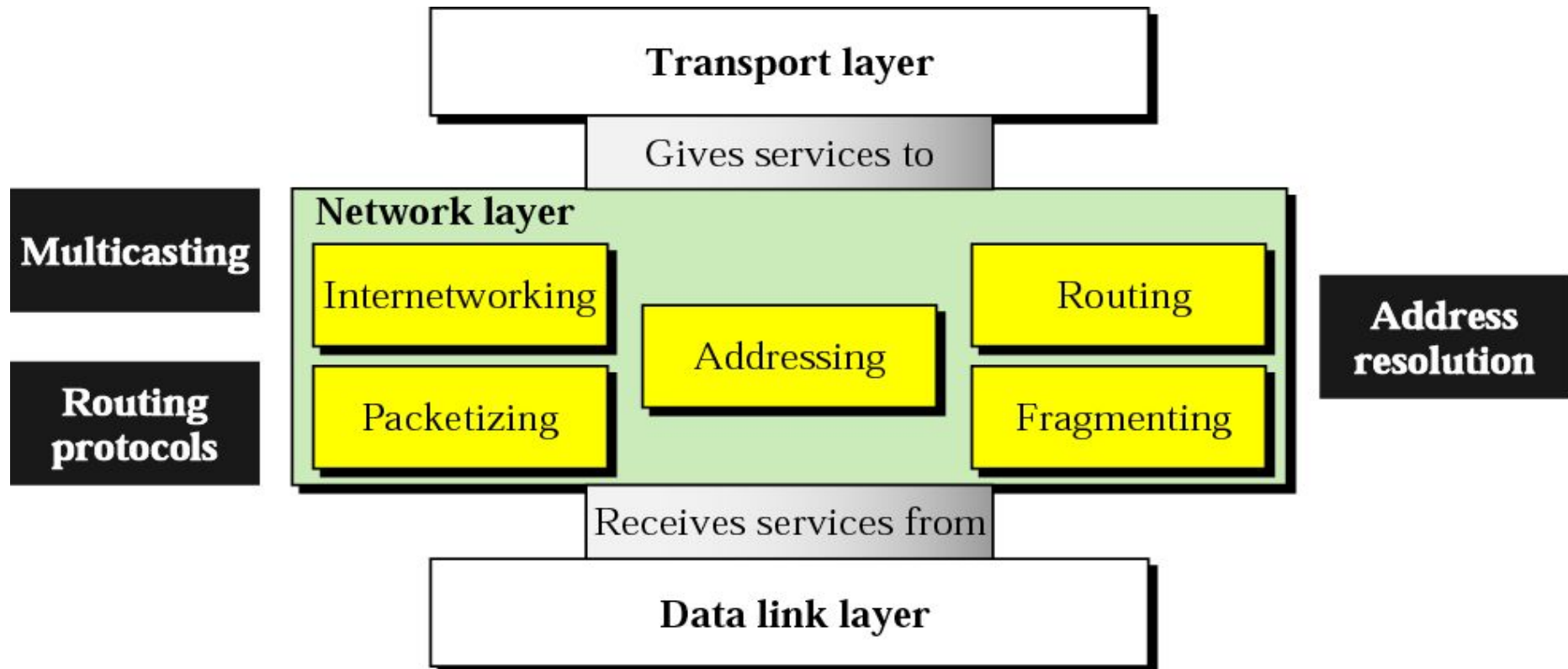


Computer Communication & Networks

Lecture 21 & 22

Network Layer: Logical Addressing

Network Layer



Network Layer Topics to Cover

Logical Addressing

Internet Protocol

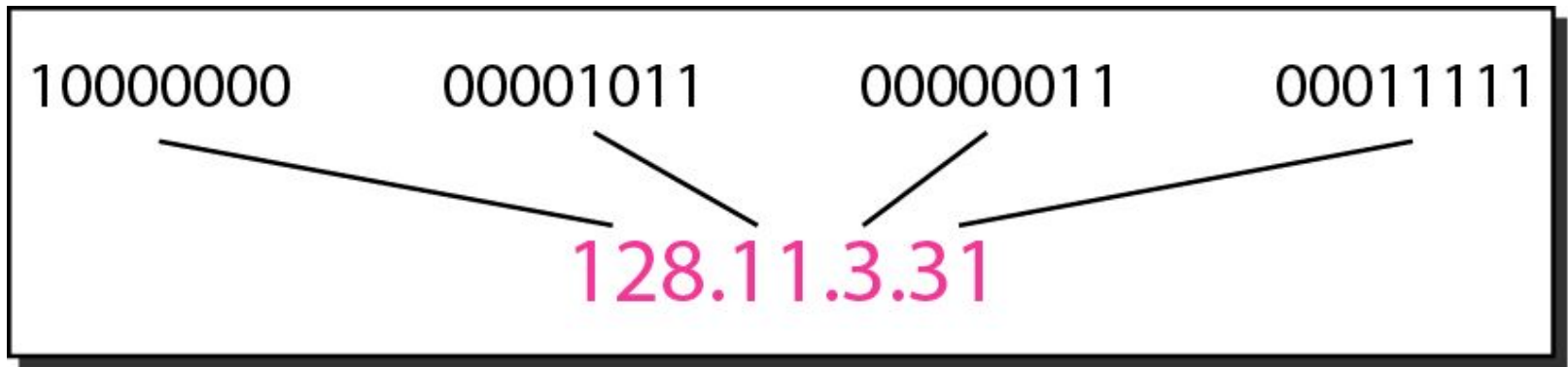
Address Mapping

Delivery, Forwarding, Routing

IPv4 Addresses

- An **IPv4 address** is a **32-bit** address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.
- The address space of IPv4 is 2^{32} or 4,294,967,296.

Dotted-decimal Notation and Binary notation for an IPv4 address



Example

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation.

a. 129.11.11.239

b. 193.131.27.255

Example

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 221.34.7.82

Solution

We replace each decimal number with its binary equivalent (see Appendix B).

a. 01101111 00111000 00101101 01001110

b. 11011101 00100010 00000111 01010010

Example

Find the error, if any, in the following IPv4 addresses.

- a. 111.56.045.78
- b. 221.34.7.8.20
- c. 75.45.301.14
- d. 11100010.23.14.67

Solution

- a. There must be no leading zero (045).*
 - b. There can be no c. Each number needs to be less than or equal to 255.*
 - d. A mixture of binary notation and dotted-decimal notation is not allowed.*
-

Note

**In classful addressing, the address space is divided into five classes:
A, B, C, D, and E.**

Finding the classes in binary and dotted-decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

Example

Find the class of each address.

- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

Solution

- a.* The first bit is 0. This is a class A address.
- b.* The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.* The first byte is 14; the class is A.
- d.* The first byte is 252; the class is E.

Number of blocks and block size in classful IPv4 addressing

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Note

In classful addressing, a large part of the available addresses were wasted.

Default masks for classful addressing

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255 .0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255 .0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255 .0	/24

Note

Classful addressing, which is almost obsolete, is replaced with classless addressing.

Subnet Addressing

- Subnet addressing introduces another hierarchical level
- Transparent to remote networks
- Simplifies management of multiplicity of LANs
- Masking used to find subnet number

Original
address

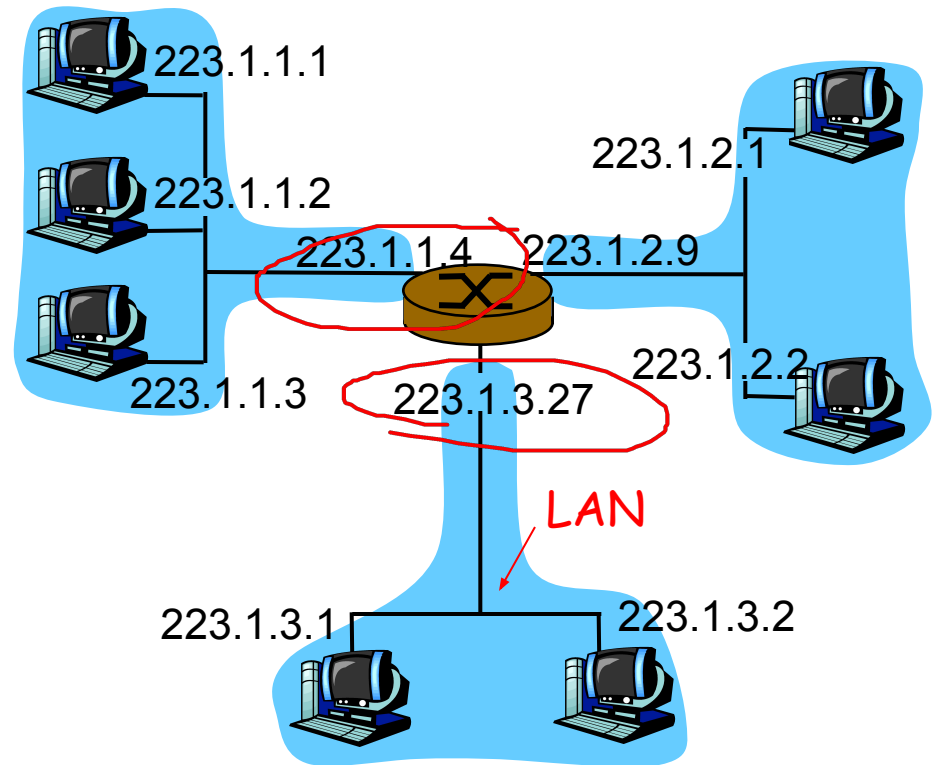
1	0	Net ID	Host ID
---	---	--------	---------

Subnetted
address

1	0	Net ID	Subnet ID	Host ID
---	---	--------	-----------	---------

Subnets

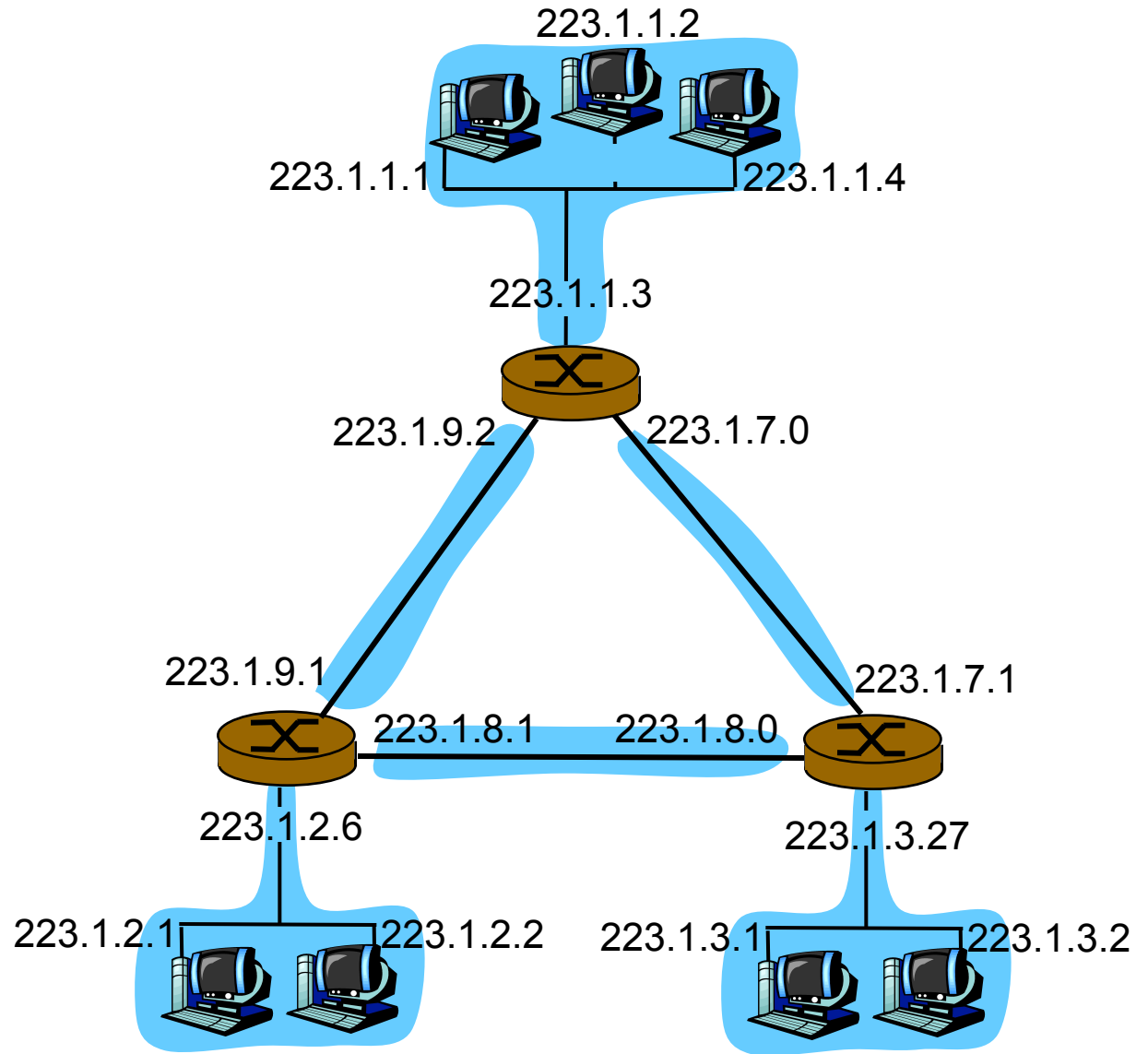
- **IP address:**
 - ❑ subnet part (high order bits)
 - ❑ host part (low order bits)
- *What's a subnet ?*
 - ❑ device interfaces with same subnet part of IP address
 - ❑ can physically reach each other without intervening router



network consisting of 3 subnets

Subnets

How many?



Subnetting Example

- Organization has Class B address (16 host ID bits) with network ID: 150.100.0.0
- Create subnets with up to 100 hosts each
 - ❑ 7 bits sufficient for each subnet
 - ❑ $16-7=9$ bits for subnet ID
- Apply subnet mask to IP addresses to find corresponding subnet
 - ❑ Example: Find subnet for 150.100.12.176
 - ❑ IP add = 10010110 01100100 00001100 10110000
 - ❑ Mask = 11111111 11111111 11111111 10000000
 - ❑ AND = 10010110 01100100 00001100 10000000
 - ❑ Subnet = 150.100.12.128
 - ❑ Subnet address used by routers within organization

Supernetting (Address Aggregation)

- Allows several networks to be combined to create a larger range of addresses.
- For example this allows an organisation to combine several Class C blocks together.
- It aggregates a contiguous block of addresses to form a single **larger** network.
- Identified by **less** bits of 1s (i.e. shorter network part) in the mask than default one.

Supernetting

- Summarize a contiguous group of class C addresses using variable-length mask

- Example: 200.158.16.0/20

IP Address (200.158.16.0) & mask length (20)

IP add = 11001000 10011110 00010000 00000000

Mask = 11111111 11111111 11110000 00000000

Contains 16 Class C blocks:

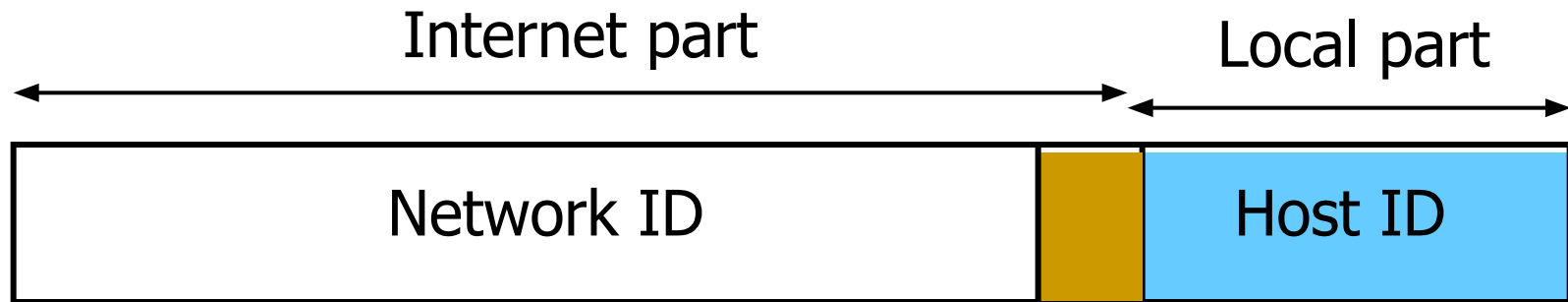
From 11001000 10011110 00010000 00000000

i.e. 200.158.16.0

Up to 11001000 10011110 00011111 00000000

i.e. 200.158.31.0

Supernet : an example



11000000 11100110 00010100 00000000
11000000 11100110 00010101 00000000
11000000 11100110 00010110 00000000
11000000 11100110 00010111 00000000

4 Class C IP network addresses

Supernet : an example

- Four class C networks were put together to form the supernet:

- ❑ 192.230.20. 0

11000000 11100110 01010000 00000000

- ❑ 192.230.21. 0

11000000 11100110 01010001 00000000

- ❑ 192.230.22. 0

11000000 11100110 01010010 00000000

- ❑ 192.230.23. 0

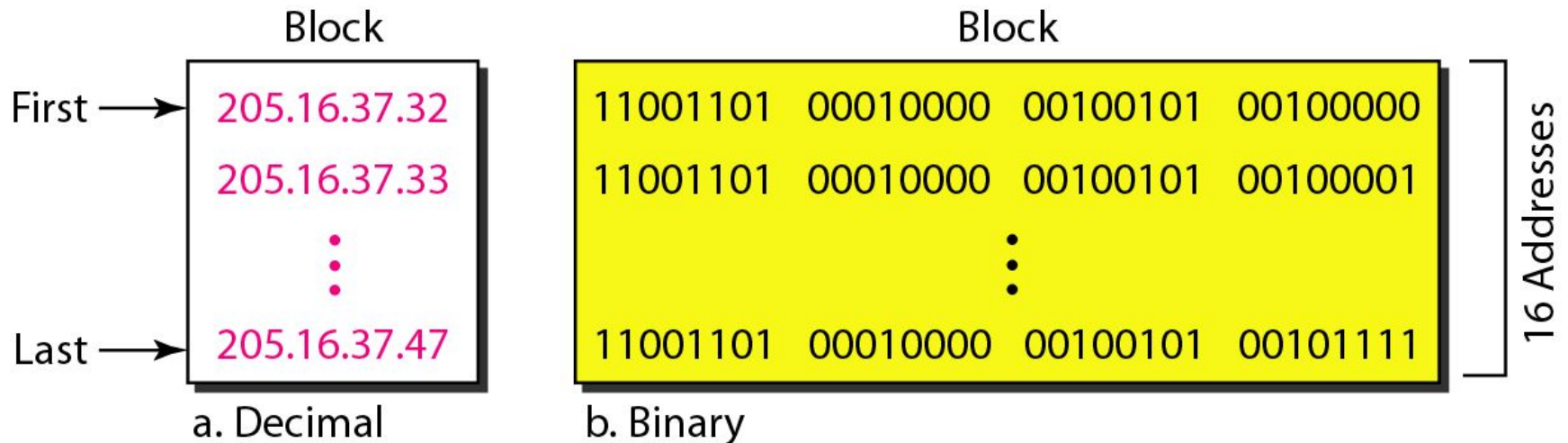
11000000 11100110 01010011 00000000

- The address of lowest network together with the supernet mask is required in the global routers for routing : 192.230.20.0 / 22

Classless Addressing

Example

Figure below shows a block of addresses, in both binary and dotted-decimal notation, granted to a small business that needs 16 addresses.



Note

In IPv4 addressing, a block of addresses can be defined as

$x.y.z.t /n$

in which $x.y.z.t$ defines one of the addresses and the $/n$ defines the mask.

Note

The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.

Example

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

Note

The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.

Example

Find the last address for the block in last Example.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

11001101 00010000 00100101 00101111

or

205.16.37.47

Note

**The number of addresses in the block
can be found by using the formula
 2^{32-n} .**

Example

Find the number of addresses in last Example.

Solution

The value of n is 28, which means that number of addresses is 2^{32-28} or 16.

Example

Another way to find the first address, the last address, and the number of addresses is to represent the mask as a 32-bit binary (or 8-digit hexadecimal) number. This is particularly useful when we are writing a program to find these pieces of information. In Last Example the /28 can be represented as

11111111 11111111 11111111 11110000

(twenty-eight 1s and four 0s).

Find

- a. The first address
- b. The last address
- c. The number of addresses.

Example (continued)

Solution

- a. The first address can be found by ANDing the given addresses with the mask. ANDing here is done bit by bit. The result of ANDing 2 bits is 1 if both bits are 1s; the result is 0 otherwise.

Address:	11001101	00010000	00100101	00100111
Mask:	11111111	11111111	11111111	11110000
First address:	11001101	00010000	00100101	00100000

Example (continued)

- b. The last address can be found by ORing the given addresses with the complement of the mask. ORing here is done bit by bit. The result of ORing 2 bits is 0 if both bits are 0s; the result is 1 otherwise. The complement of a number is found by changing each 1 to 0 and each 0 to 1.

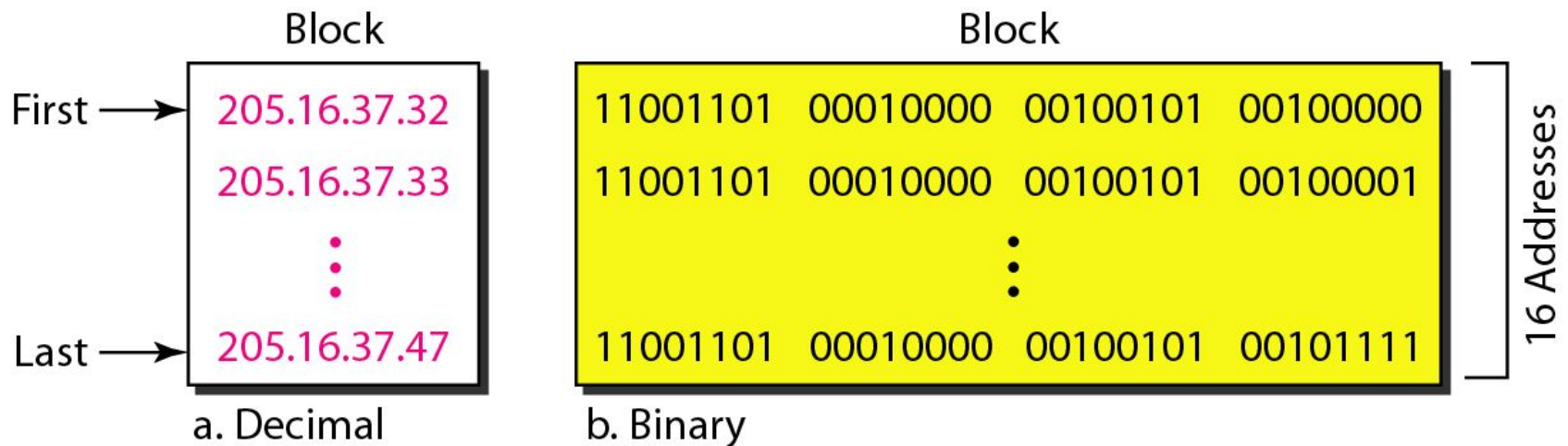
Address:	11001101	00010000	00100101	00100111
Mask complement:	00000000	00000000	00000000	00001111
Last address:	11001101	00010000	00100101	00101111

Example (Contd.)

- c. The number of addresses can be found by complementing the mask, interpreting it as a decimal number, and adding 1 to it.

Mask complement:	00000000	00000000	00000000	00001111
Number of addresses:	$15 + 1 = 16$			

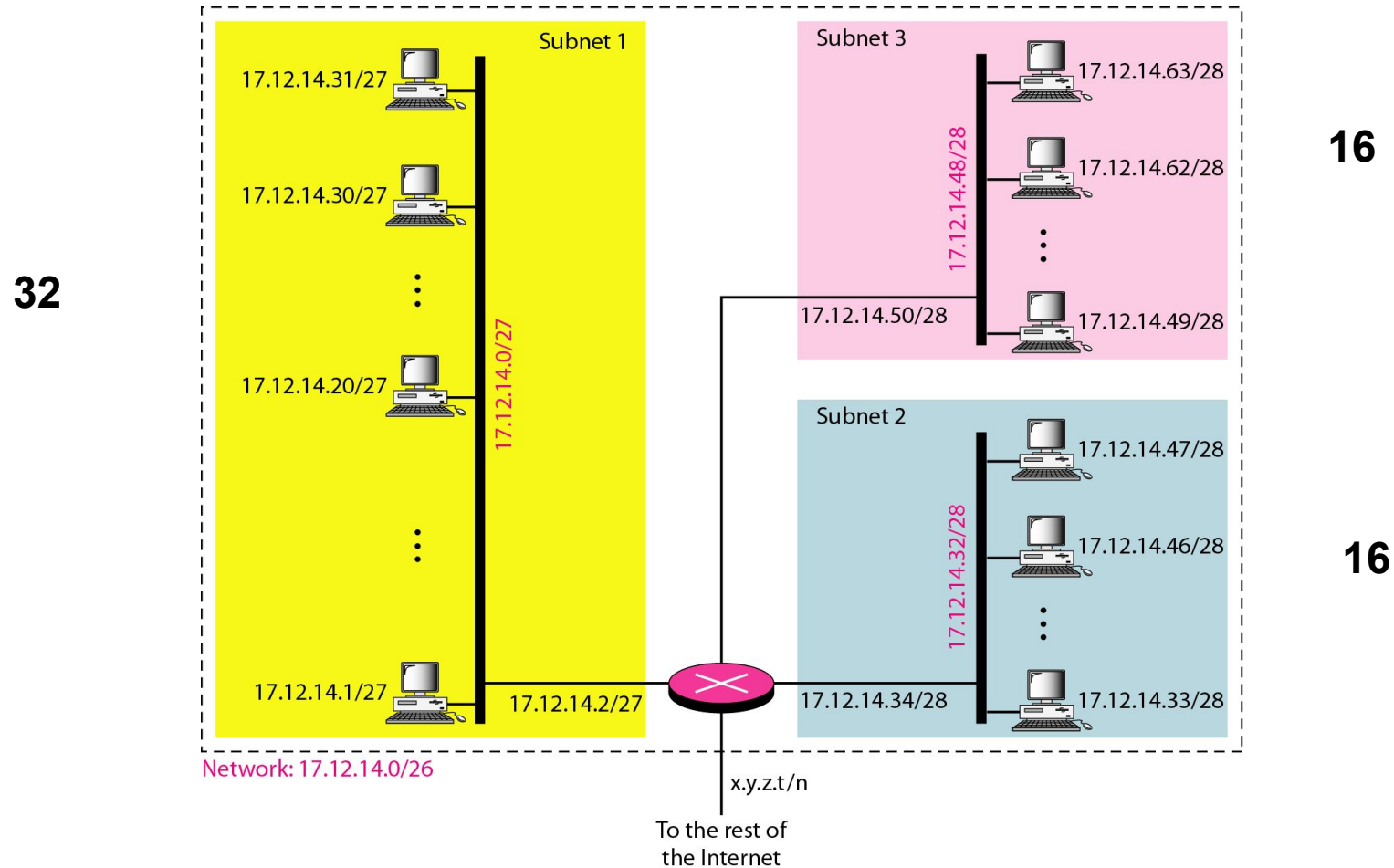
A network configuration for the block 205.16.37.32/28



Note

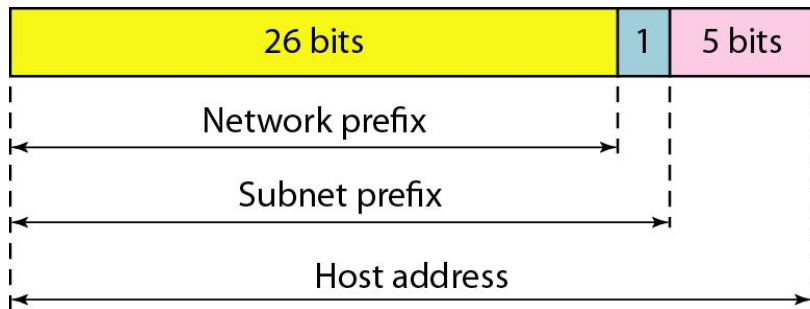
The first address in a block is normally not assigned to any device; it is used as the network address that represents the organization to the rest of the world.

Configuration and Addresses in a Subnetted Network

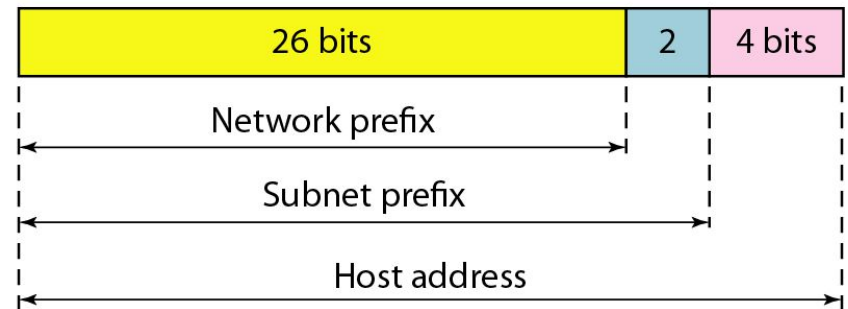


Three-level hierarchy in an IPv4 address

Subnet 1



Subnets 2 and 3



Example

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- a. The first group has 64 customers; each needs 256 addresses.
- b. The second group has 128 customers; each needs 128 addresses.
- c. The third group has 128 customers; each needs 64 addresses.

Design the subblocks and find out how many addresses are still available after these allocations.

Example (continued)

Solution

Group 1

For this group, each customer needs 256 addresses. This means that 8 ($\log_2 256$) bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		

Example (continued)

Group 2

For this group, each customer needs 128 addresses. This means that 7 ($\log_2 128$) bits are needed to define each host. The prefix length is then $32 - 7 = 25$. The addresses are

<i>1st Customer:</i>	<i>190.100.64.0/25</i>	<i>190.100.64.127/25</i>
<i>2nd Customer:</i>	<i>190.100.64.128/25</i>	<i>190.100.64.255/25</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.127.128/25</i>	<i>190.100.127.255/25</i>
<i>Total = $128 \times 128 = 16,384$</i>		

Example (continued)

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then $32 - 6 = 26$. The addresses are

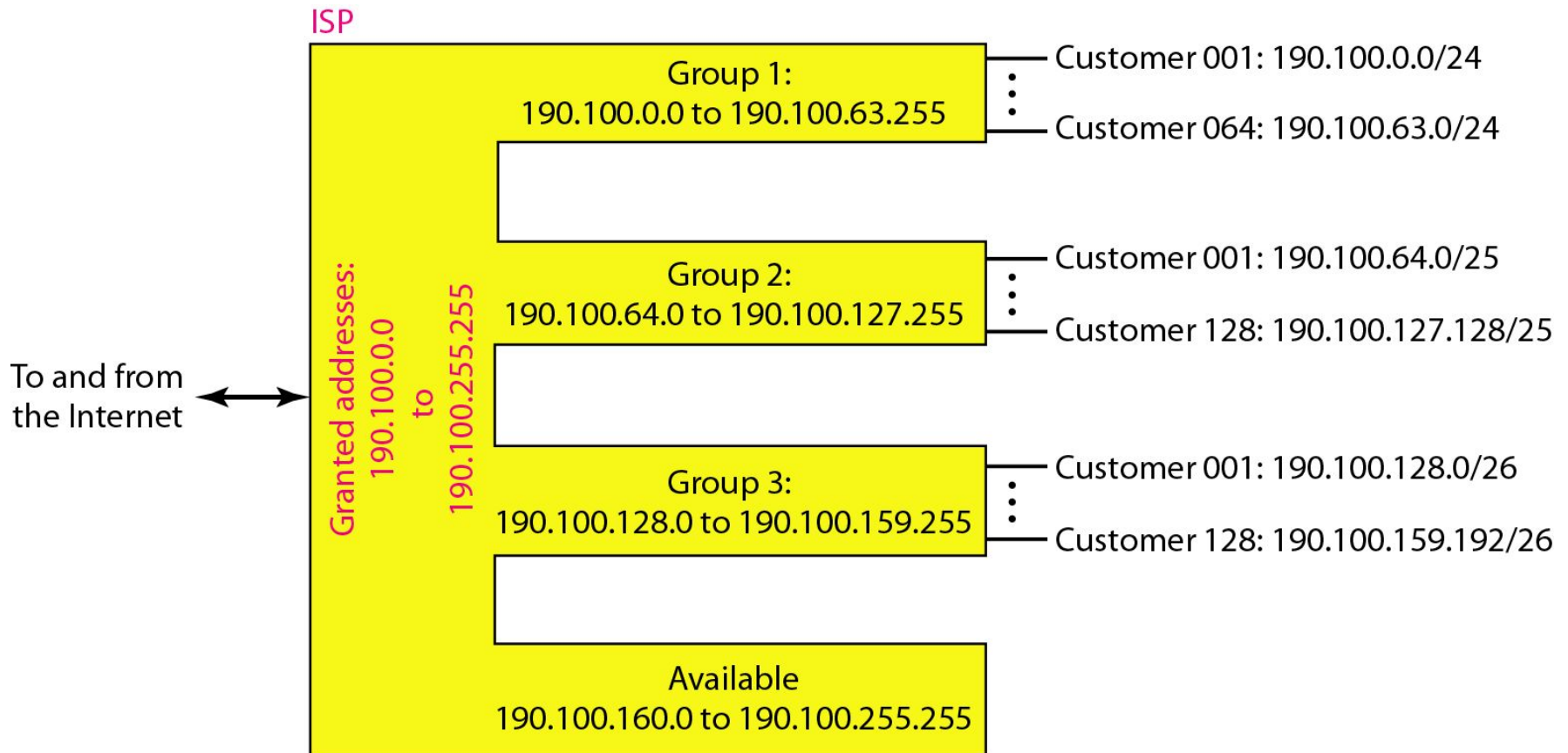
<i>1st Customer:</i>	<i>190.100.128.0/26</i>	<i>190.100.128.63/26</i>
<i>2nd Customer:</i>	<i>190.100.128.64/26</i>	<i>190.100.128.127/26</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.159.192/26</i>	<i>190.100.159.255/26</i>
<i>Total = $128 \times 64 = 8192$</i>		

Number of granted addresses to the ISP: 65,536

Number of allocated addresses by the ISP: 40,960

Number of available addresses: 24,576

An example of address allocation and distribution by an ISP



Example

- An organization has a class C network: 200.1.1.0, and it wants to form subnets for 4 departments with the number of hosts as follows:
- Subnet A: 72 hosts
- Subnet B: 35 hosts
- Subnet C: 20 hosts
- Subnet D: 18 hosts
- There are 145 hosts in total.
- Provide a possible arrangement of the network address space, together with the respective range of IP addresses for each subnet. Explain your work!